

Dynamic Pricing Strategy for Ride-Sharing Services

Dataset: [dynamic_pricing.csv](#)

A ride-sharing company wants to optimize ride fares based on real-time market conditions instead of using fixed pricing rules. Students will analyze demand, supply, booking time, and ride characteristics to understand factors affecting ride prices and build a predictive pricing model. Expected insights include identifying surge-pricing conditions, understanding demand-supply imbalance, and evaluating factors that most strongly influence ride fares.

Final Project Deliverables:

1. Data cleaning report
2. Exploratory Data Analysis (EDA) report
3. Demand and supply analysis
4. Pricing factor analysis
5. Visualization dashboard
6. Ride fare prediction model
7. Business and marketing recommendations
8. Executive summary

Dataset Description:

Variable	Description
Number_of_Riders	Number of customers requesting rides during a given period or scenario, representing ride demand.
Number_of_Drivers	Number of available drivers during the same period, representing ride supply.
Location_Category	Geographic category where the ride is requested (e.g., Urban, Suburban, Rural).
Customer_Loyalty_Statu s	Loyalty tier of the customer (e.g., Regular, Silver, Gold).

Number_of_Past_Rides	Total number of rides previously completed by the customer, indicating usage history.
Average_Ratings	Average customer rating score based on previous rides.
Time_of_Booking	Time period when the ride was booked (e.g., Morning, Afternoon, Evening, Night).
Vehicle_Type	Type of vehicle selected for the ride (e.g., Economy, Premium).
Expected_Ride_Duration	Estimated duration of the ride, typically measured in minutes.
Historical_Cost_of_Ride	Historical fare charged for the ride. This is the target variable for pricing prediction analyses.